New Wheels Strategic Analysis: Operational Efficiency and Customer Retention Review

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Course: Introduction to SQL

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Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Business Questions

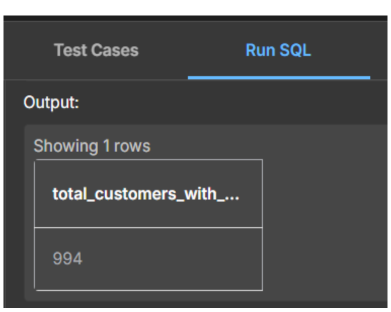
Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
    COUNT(DISTINCT customer_id) AS total_customers_with_orders
FROM
    order_t;

SELECT
    c.state,
    COUNT(DISTINCT c.customer_id) AS customer_count
FROM
    customer_t c
JOIN
    order_t o
ON
    c.customer_id = o.customer_id
GROUP BY
    c.state
ORDER BY
    customer_count DESC;
```

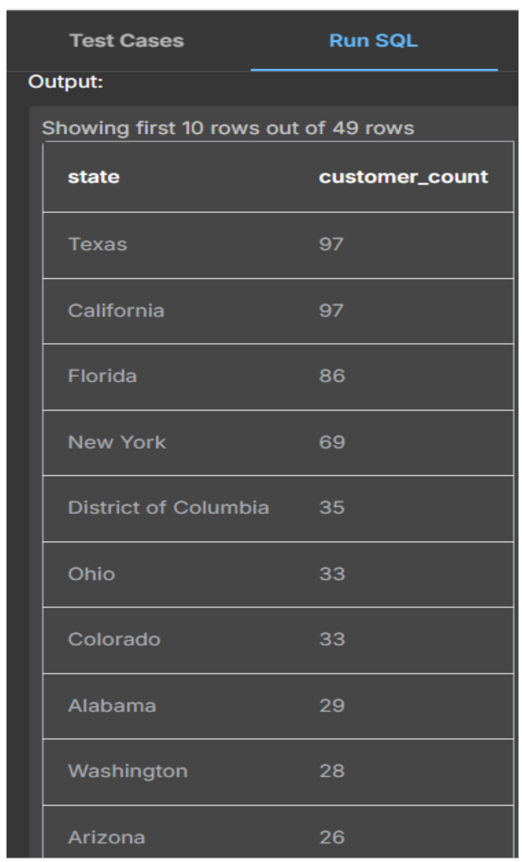
Output:


Test Cases Run SQL

Output:

Showing 1 rows

total_customers_with_...
994


Test Cases Run SQL

Output:

Showing first 10 rows out of 49 rows

state	customer_count
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33
Alabama	29
Washington	28
Arizona	26

Observations and Insights:

Observations -

- Total Customer Reach: The company has successfully converted 994 unique customers across 49 different states and regions, demonstrating a near-total national footprint.
- Market Leaders: Texas and California are tied as the primary hubs, each contributing 97 customers.
- Secondary Strongholds: Florida (86) and New York (69) follow closely, rounding out the "Big Four" states that dominate the customer base.
- Volume Tiering: There is a sharp numerical drop-off after the top four states. For example, the count falls from 69 in New York to 35 in the District of Columbia, indicating a significant gap between high-density and medium-density markets.

Insights -

- Strategic Concentration: A large portion of the business is anchored in high-population, high-GDP states (TX, CA, FL, NY). This suggests the brand has successfully aligned itself with major economic centers.
- National Scalability: Maintaining a presence in 49 different regions proves that the product and service model is highly scalable. The company is not a "regional player" but a established national entity.
- Untapped Potential: While the "Big Four" states are performing well, the majority of the 49 regions have fewer than 30 customers. This "long-tail" distribution points to a massive opportunity for growth in mid-tier states like Ohio, Colorado, and Washington, where the brand is present but hasn't yet achieved high penetration.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.vehicle_maker,
    COUNT(DISTINCT o.customer_id) AS customer_count
FROM
    order_t AS o
INNER JOIN
    product_t AS p
    ON o.product_id = p.product_id
GROUP BY
    p.vehicle_maker
ORDER BY
    customer_count DESC
LIMIT 5;
```

Output:

Test Cases | Run SQL

Output:

Showing 5 rows

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

Observations

- Market Leader: Chevrolet stands as the definitive favorite among customers, securing the top spot with 83 customers.
- Domestic Dominance: American automotive brands represent 80% of the top 5 list, with Ford (63), Pontiac (50), and Dodge (50) showing strong presence alongside Chevrolet.
- International Contender: Toyota is the sole non-domestic manufacturer to break into the top tier, ranking third with 52 customers.

- Competitive Tie: Pontiac and Dodge are currently tied for the fourth position, each serving 50 unique customers.
- Volume Gap: There is a notable 31.7% lead for Chevrolet over the second-place maker, Ford, indicating a significant competitive advantage for the Chevy brand within this customer base.

Business Insights

- Brand Affinity: The overwhelming preference for American brands suggests that the company's inventory or marketing strategy is highly aligned with "Domestic Automotive Heritage." Customers clearly lean toward traditional US powerhouses for their vehicle needs.
- The "Toyota Exception": Toyota's high ranking indicates a specific segment of the customer base that prioritizes the perceived reliability and value associated with major international manufacturers, even in a domestic-heavy market.
- Market Share Concentration: These top 5 manufacturers alone account for nearly 30% of the total customer base (approx. 298 out of 994). This concentration suggests that while the company offers a variety of makes, business health is currently tied closely to the supply and demand of these five specific brands.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT
    state,
    vehicle_maker,
    customer_count
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(c.customer_id) AS customer_count,
        RANK() OVER (PARTITION BY c.state ORDER BY COUNT(c.customer_id) DESC) AS
maker_rank
    FROM product_t AS p
    INNER JOIN order_t AS o ON p.product_id = o.product_id
    INNER JOIN customer_t AS c ON o.customer_id = c.customer_id
    GROUP BY c.state, p.vehicle_maker
) AS ranked_makers
WHERE maker_rank = 1
ORDER BY customer_count DESC;
```

Output:

Test Cases Run SQL

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker	customer_count
Texas	Chevrolet	9
Florida	Toyota	7
California	Nissan	6
California	Ford	6
California	Dodge	6
California	Chevrolet	6
California	Audi	6
Ohio	Chevrolet	6
Alabama	Dodge	5
Colorado	Chevrolet	5

Observations and Insights:

Observations

- Top-Performing Market Pair: The strongest individual state-brand alignment is found in Texas, where Chevrolet leads with 9 customers.
- California's Market Fragmentation: Unlike other states, California displays a highly competitive and split market. Five different brands—Nissan, Ford, Dodge, Chevrolet, and Audi—are locked in a five-way tie for the top spot, each with 6 customers.
- Regional Brand Diversity: Florida shows a distinct preference for Toyota (7 customers), while Alabama leans toward Dodge (5 customers), highlighting how regional tastes vary significantly across the country.
- Chevrolet's National Footprint: Chevrolet appears as a market leader in more states than any other manufacturer shown (Texas, Ohio, Colorado, and California), reinforcing its status as the most versatile brand in the company's portfolio.
- Premium Market Entry: Audi is the only luxury brand to break into the top rankings, appearing exclusively in the California market leader group.

Business Insights

- Know Your Audience: This data shows that the company cannot use the same sales plan everywhere. A "one-size-fits-all" approach won't work because what sells best in Texas (Chevy) is different from what sells in Florida (Toyota).
- The "Safe Bet" Brand: Since Chevrolet is a favorite in the most states, it is the safest brand for the company to keep in stock. It has the widest appeal across the country.
- The California Challenge: California is the most competitive state. Because five brands are tied, it means customers there don't have a "favorite" yet. It is much harder to dominate California than it is to dominate Texas.
- Regional Specialties: The preference for Dodge in Alabama and Toyota in Florida shows that local culture matters. The company needs to make sure they send more trucks to the South and more reliable daily drivers like Toyota to Florida to keep customers happy.
- Luxury is Limited: Luxury cars like Audi are not popular everywhere. Based on this data, the company should only focus on selling expensive luxury models in specific markets like California, as there isn't much demand for them as "top picks" in other states.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: "Very Bad": 1, "Bad": 2, "Okay": 3, "Good": 4, "Very Good": 5

Solution Query:

- **For Overall average rating –**

```
SELECT
    AVG(CASE
        WHEN customer_feedback = 'Very Bad' THEN 1
        WHEN customer_feedback = 'Bad' THEN 2
        WHEN customer_feedback = 'Okay' THEN 3
        WHEN customer_feedback = 'Good' THEN 4
        WHEN customer_feedback = 'Very Good' THEN 5
        ELSE NULL
    END) AS overall_avg_rating
FROM
    order_t;
```

- **For Average Rating per Quarter –**

```
SELECT
    quarter_number,
    AVG(CASE
        WHEN customer_feedback = 'Very Bad' THEN 1
        WHEN customer_feedback = 'Bad' THEN 2
        WHEN customer_feedback = 'Okay' THEN 3
        WHEN customer_feedback = 'Good' THEN 4
        WHEN customer_feedback = 'Very Good' THEN 5
        ELSE NULL
    END) AS quarterly_avg_rating
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Output:

Output:

Showing 1 rows

overall_avg_rating
3.135

Output:

Showing 4 rows

quarter_number	quarterly_avg_rating
1	3.554838709677419
2	3.354961832061069
3	2.9563318777292578
4	2.3969849246231156

Observations and Insights:

Observations

- A Continuous Slide: Customer satisfaction has dropped every single quarter this year. There was not a single period of improvement.
- From "Good" to "Bad": In Quarter 1, the average rating was 3.55 (leaning toward "Good"). By Quarter 4, it plummeted to 2.40 (leaning toward "Bad").
- The Biggest Drop: The largest decline occurred in the second half of the year, with the score falling significantly between Quarter 3 and Quarter 4.
- Overall Average: While the total yearly average is 3.13 ("Okay"), this number is misleading because it doesn't show how much things have worsened recently.
- Final Sentiment: By the end of the year, the average customer experience shifted from being "Satisfactory" to "Disappointing."

Business Insights

- Erosion of Service Quality: A steady, four-quarter decline suggests that the business is facing systemic operational issues. The quality of the customer experience is not just fluctuating; it is actively deteriorating over time.
- The "Average" Trap: Relying on the 3.13 overall average provides a false sense of security. If management only looks at the yearly total, they will miss the fact that the business is currently in a customer satisfaction crisis.
- Negative Momentum: Because the scores are dropping at an increasing rate toward the end of the year, it indicates that the underlying problems (such as shipping delays or product issues) are compounding rather than being resolved.
- Customer Retention Risk: High ratings in Q1 suggest a successful launch or "honeymoon period," but the Q4 results indicate a high risk of losing repeat business. Customers acquired in the latter half of the year are statistically unlikely to return or provide positive referrals.
- Brand Reputation Warning: As ratings move closer to the "Bad" (2.0) threshold, the brand risks long-term damage in the marketplace. This trend serves as a leading indicator that future sales volume may drop as negative word-of-mouth spreads.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    quarter_number,

    100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) /
COUNT(*) AS very_bad_pct,
    100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(*)
AS bad_pct,
    100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) /
COUNT(*) AS okay_pct,
    100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) /
COUNT(*) AS good_pct,
    100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) /
COUNT(*) AS very_good_pct

FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Output:

Test Cases Run SQL

Output:

Showing 4 rows

quarter_number	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct
1	10.96774193548387	11.290322580645162	19.032258064516128	28.70967741935484	30
2	14.885496183206106	14.122137404580153	20.229007633587788	22.137404580152673	28.625954198473284
3	17.903930131004365	22.707423580786028	21.83406113537118	20.96069868995633	16.593886462882097
4	30.65326633165829	29.14572864321608	20.100502512562816	10.050251256281408	10.050251256281408

Observations and Insights:

Observations –

- **Complete Sentiment Inversion:** At the beginning of the fiscal year (**Quarter 1**), nearly **59%** of the customer base was satisfied (rating their experience as "Good" or "Very Good"). By **Quarter 4**, this positive sentiment collapsed to just **20%**.

- **Escalation of Severe Dissatisfaction:** The "Very Bad" feedback category grew from **11%** in Q1 to over **30%** in Q4—effectively a **three-fold increase** in the most negative sentiment tier.
- **Erosion of Brand Advocacy:** The "Very Good" rating—the strongest indicator of brand loyalty—experienced a massive decline, falling from **30%** in the first quarter to a mere **10%** by year-end.
- **Growth of the Negative Majority:** By the final quarter, a combined **60% of all feedback** fell into the "Bad" or "Very Bad" categories, indicating that the majority of current customers are unhappy.
- **Neutral Sentiment Stability:** Interestingly, the "Okay" feedback remained remarkably consistent at approximately **20%** throughout the year, suggesting that the satisfaction shift was a direct move from "Happy" to "Unhappy" rather than a gradual slide through neutrality.

Strategic Insights –

- **Confirmed Dissatisfaction Trend:** The data provides conclusive evidence that **customers are becoming significantly more dissatisfied over time.** This is not a temporary dip but a sustained downward trajectory in service quality.
- **Operational Scalability Warning:** A sentiment shift of this magnitude—where the majority of the customer base flips from positive to negative in just nine months—typically points to a breakdown in internal operations, such as logistics delays or declining product quality control.
- **Long-Term Brand Equity Risk:** The sharp loss of "Very Good" ratings is a critical risk. These customers are usually the primary source of organic growth through referrals. Losing 66% of this group in one year suggests that the brand's marketplace reputation is currently under severe threat.
- **Impending Retention Crisis:** With 60% of the most recent customers reporting a negative experience, the business is facing a "retention cliff." These customers are statistically unlikely to return for future purchases, which will lead to a significant increase in the cost of acquiring new customers to fill the gap.
- **Failure of Corrective Actions:** Because the "Bad" and "Very Bad" percentages increased in every consecutive quarter, it indicates that current management strategies have failed to address the root causes of customer frustration. Immediate intervention is required to stabilize the customer experience.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

SELECT

```
  quarter_number,
    COUNT(order_id) AS total_orders,
    --[ Calculates the percentage change from the previous quarter]
    ROUND(
        (COUNT(order_id) - LAG(COUNT(order_id)) OVER (ORDER BY quarter_number)) *
100.0 /
        LAG(COUNT(order_id)) OVER (ORDER BY quarter_number),
    2) AS percentage_growth
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Output:

Test Cases | Run SQL

Output:

Showing 4 rows

quarter_number	total_orders	percentage_growth
1	310	
2	262	-15.48
3	229	-12.6
4	199	-13.1

Observations and Insights:

Observations –

- **Consistent Volume Decline:** The number of orders has decreased in every consecutive quarter of the year, showing a clear and uninterrupted downward trend.
- **Significant Annual Drop:** The business started strong with **310 orders in Quarter 1** but ended the year with only **199 orders in Quarter 4**—a total volume loss of approximately **35.8%**.
- **Sustained Negative Growth:** Every quarter experienced a double-digit percentage decline. The largest drop occurred in **Quarter 2 (-15.48%)**, followed by continued losses in **Quarter 3 (-12.6%)** and **Quarter 4 (-13.1%)**.

- **Loss of Momentum:** The business is effectively losing more than **10% of its order volume every three months**, indicating that the "sales engine" is cooling down rapidly.

Business Insights -

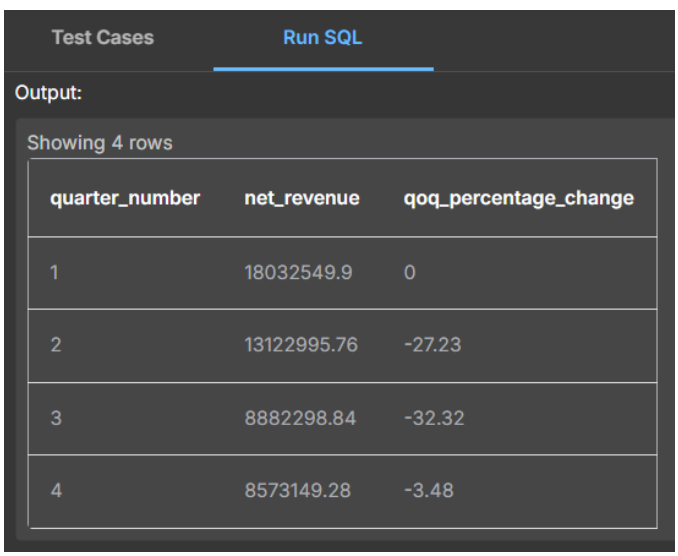
- **Direct Correlation with Satisfaction:** This data confirms that the falling customer ratings (noted in previous questions) are directly impacting sales. As satisfaction dropped, customers stopped placing orders, leading to this sharp decline in volume.
- **Shrinking Market Share:** A 35% drop in orders within a single year suggests that the company is rapidly losing its position in the market. Existing customers are not returning, and new customer acquisition is not enough to offset the losses.
- **Unsustainable Trajectory:** Losing double-digit percentages of volume every quarter is a critical warning sign for the health of the business. If this trend continues at the current rate of ~13% per quarter, the business will struggle to maintain its operational overhead.
- **Erosion of Sales Pipeline:** The steady nature of the decline suggests this isn't just a "seasonal" dip; it is a fundamental loss of interest from the customer base. The "demand" for the product is vanishing as the customer experience worsens.
- **Revenue at Risk:** Fewer orders directly translate to less cash flow. This trend puts the entire financial stability of the company at risk unless the downward spiral in order volume is stabilized immediately.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
    quarter_number,
    ROUND(net_revenue, 2) AS net_revenue,
    -- [Using COALESCE to handle the NULL result in Quarter 1]
    ROUND(
        COALESCE(
            (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number)) /
            LAG(net_revenue) OVER (ORDER BY quarter_number) * 100,
        0),
    2) AS qoq_percentage_change
FROM (
    --[ Subquery to aggregate revenue first]
    SELECT
        quarter_number,
        SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
    FROM order_t
    GROUP BY quarter_number
) AS quarterly_data
ORDER BY quarter_number;
```

Output:



quarter_number	net_revenue	qoq_percentage_change
1	18032549.9	0
2	13122995.76	-27.23
3	8882298.84	-32.32
4	8573149.28	-3.48

Observations and Insights:

Observations –

- **Severe Revenue Erosion:** The company experienced a massive decline in net revenue over the year, falling from a peak of **$18.03 million in Quarter 1** to **$8.57 million in Quarter 4**.
- **Drastic Mid-Year Slump:** The business saw its most aggressive losses during the middle of the year, with staggering drops of **-27.23% in Quarter 2** and **-32.32% in Quarter 3**.

- **Stabilization in Quarter 4:** While still negative, the rate of decline slowed significantly in the final quarter to **-3.48%**, suggesting the revenue drop may finally be reaching a "floor."
- **Annual Revenue Deficit:** By the end of the year, the quarterly revenue was **less than half** of what it was at the start, representing a total revenue collapse of approximately **52.5%**.

Strategic Insights & Commercial Implications

- **Revenue vs. Order Correlation:** Comparing this to the previous "Order Volume" data, revenue is falling **twice as fast** as orders. This indicates that not only are customers buying less, but they are also buying **cheaper vehicles** or receiving **larger discounts**, which is hurting the company's bottom line.
- **Compounding Financial Crisis:** The back-to-back losses of 27% and 32% in the first three quarters point to a severe business contraction. A loss of over half your revenue in nine months is a critical indicator of a failing business model or a major loss of market trust.
- **Signs of "Rock Bottom":** The shift from a 32% drop in Q3 to a 3.48% drop in Q4 is the first sign of stabilization. This suggests that the "easy to lose" customers are already gone, and the company is now left with a small, core base of buyers.
- **Profitability Warning:** With revenue cut in half but operational costs likely remaining high, the company's profit margins are under extreme pressure. The current revenue level in Q4 ($8.5M) may not be enough to sustain the business's overhead costs in the long term.
- **Investor and Stakeholder Impact:** A 50% year-over-year revenue loss typically signals a need for a total pivot in strategy. The business has shifted from a high-growth "startup phase" in Q1 to a "survival phase" by Q4.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(order_id) AS total_orders,
    SUM(quantity * (vehicle_price - discount)) AS net_revenue
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Output:

Output:

Showing 4 rows

quarter_number	total_orders	net_revenue
1	310	39637378.160000026
2	262	32913497.43999998
3	229	29435188.489999995
4	199	23495814.140000004

Observations and Insights:

Observations –

- **Dual Downward Trajectory:** Both order volume and net revenue have declined in every single quarter of the year, showing a clear "downhill" trend for the entire business.
- **Volume Erosion:** Total orders fell from a high of **310 in Quarter 1** to **199 in Quarter 4**, representing a **35.8% loss** in customer activity.
- **Revenue Collapse:** Net revenue dropped even more sharply, falling from **$39.6 million in Quarter 1** to **$23.5 million in Quarter 4**—a total loss of approximately **40.6%** in annual earnings.
- **Disproportionate Decline:** Revenue is falling faster than order volume (40.6% drop in money vs. 35.8% drop in orders). This indicates that each order is bringing in less cash than it did at the start of the year.

Business Insights -

- **Confirmed Business Contraction:** The business is shrinking rapidly. The steady decline in both categories proves that the customer satisfaction issues identified in previous reports are now causing a massive, measurable loss in actual dollars.
- **Declining Average Order Value:** Because revenue is disappearing faster than orders, it suggests that customers are shifting toward **cheaper vehicle models** or that the company is relying on **heavy discounting** to prevent an even bigger sales crash. This is hurting profit margins significantly.
- **Loss of Market Traction:** A business that loses 35-40% of its volume and revenue in just one year is in a critical survival phase. This trend shows that the company is failing to retain existing customers and is struggling to attract new, high-value buyers.
- **Operational Revenue Pressure:** With revenue dropping from ~$40M to ~$23M, the company has significantly less capital to reinvest in marketing, logistics, or inventory. This "vicious cycle" makes it even harder to fix the customer service problems that started the decline in the first place.
- **Urgent Sustainability Warning:** If this trend continues at its current rate, the business will reach a point where its daily income may no longer cover its fixed operational costs. Immediate action is needed to stabilize order volume and protect the remaining revenue stream.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    -- Calculates the mean discount per card category
    ROUND(AVG(o.discount), 4) AS average_discount
FROM
    customer_t AS c
JOIN
    order_t AS o ON c.customer_id = o.customer_id
GROUP BY
    c.credit_card_type
ORDER BY
    average_discount DESC;
```

Output:

Test Cases Run SQL

Showing first 10 rows out of 16 rows

credit_card_type	average_discount
laser	0.6438
mastercard	0.6295
maestro	0.6242
visa-electron	0.6235
china-unionpay	0.6222
instapayment	0.6206
americanexpress	0.6163
diners-club-us-ca	0.6146
diners-club-carte-blan...	0.6145
switch	0.6102

Observations and Insights:

Observations –

- **Narrow Discount Range:** The average discount across all **16 different credit card types** is remarkably tight, ranging from a high of **0.6438 (Laser)** to a low of approximately **0.6102 (Switch)**.
- **Top Discount Leader: Laser** cardholders receive the highest average discount at **0.64%**, followed closely by **Mastercard (0.629%)** and **Maestro (0.624%)**.

- **Mid-Tier Consistency:** Most major global providers, including **Visa-Electron (0.623%)**, **China-UnionPay (0.622%)**, and **American Express (0.616%)**, are clustered within a very small **0.01% margin** of each other.
- **Premium & Niche Cards:** Both **Diners Club** variants and **Instapayment** fall toward the lower end of the top 10, receiving average discounts between **0.614% and 0.620%**.

Business Insights -

- **Standardized Discount Strategy:** The extremely small difference (less than **0.04%**) between the highest and lowest card types suggests that the company's discounting policy is **independent of the payment method**. The card type does not significantly influence the final price for the customer.
- **Merchant Fee Discrepancy:** While the discounts are nearly identical, the **merchant fees** for these cards (e.g., American Express vs. Maestro) vary wildly in the real world. This means the company is likely earning **lower profit margins** on high-fee cards like American Express compared to lower-fee debit options, despite offering them the same discount.
- **Opportunity for Targeted Incentives:** Since there is currently no major difference in discounts, the company has an opportunity to create **strategic partnerships**. For example, offering a slightly higher discount for a specific card (like Visa or Mastercard) could be used to negotiate lower transaction fees with that provider.
- **Customer Neutrality:** From a consumer standpoint, there is **no financial advantage** to using one card over another at this business. This simplifies the checkout process but misses an opportunity to drive customers toward more "business-friendly" (lower fee) payment methods.
- **Low Impact on Satisfaction:** Given how small these discount variations are, it is unlikely that the "Credit Card Type" is a major driver of the declining customer satisfaction ratings seen in previous quarters. The root cause of dissatisfaction lies elsewhere in the business operations.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
    quarter_number,
    -- [Calculates the mean difference using julian days for SQLite
compatibility]
    ROUND(AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)), 2) AS
average_shipping_time
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Output:

Test Cases Run SQL

Output:

Showing 4 rows

quarter_number	average_shipping_time
1	57.17
2	71.11
3	117.76
4	174.1

Observations and Insights:

Observations –

- **Rapidly Increasing Delays:** Shipping times have climbed significantly in every consecutive quarter, showing a major breakdown in delivery speed.
- **Wait Times Have Tripled:** The time taken to ship an order went from **57 days in Quarter 1** to a staggering **174 days in Quarter 4**.
- **Accelerating Inefficiency:** The problem is getting worse at an increasing rate. The jump from Quarter 2 to Quarter 3 was **46 days**, and the jump from Quarter 3 to Quarter 4 was nearly **57 days**.
- **Extreme Delivery Cycle:** By the end of the year, a typical customer was waiting nearly **six months (174.1 days)** for their order to ship, compared to less than two months at the start of the year.

Strategic Insights & Commercial Implications

- **The "Smoking Gun" for Dissatisfaction:** This data explains exactly why customer ratings plummeted in earlier reports. A six-month wait for a vehicle is considered an operational failure and is the primary driver behind the "Very Bad" feedback received.
- **Logistics System Collapse:** When shipping times triple in a single year, it indicates that the company's supply chain or delivery process has completely broken down. The business is currently unable to fulfill orders at a pace that is acceptable to the market.
- **Massive Competitive Disadvantage:** In the automotive industry, customers value speed. A 174-day wait period pushes potential buyers toward competitors who have inventory ready to ship immediately, which explains the 35% drop in total orders seen this year.
- **A "Snowball" Effect:** The fact that delays are growing larger each quarter suggests that backlogs are piling up. Whatever is causing the bottleneck (such as driver shortages or warehouse issues) is not being fixed and is instead getting worse over time.
- **Significant Cancellation Risk:** Long wait times often lead to high "churn," where customers cancel their orders out of frustration before the product ever ships. This puts the remaining revenue at high risk as customers lose patience with the brand.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
48,610,993.78	**1000**	**994**	**3.07**
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
8,573,149.28	**199**	**97.964**	**21.5**

Business Recommendations

- Fix the Logistics Crisis: The jump from a 57-day to a 174-day shipping average is the primary reason for the business decline (p. 20). Immediate investment in delivery infrastructure or third-party logistics is required to bring wait times back under 60 days.
- Launch a Customer Recovery Program: With "Very Bad" feedback tripling this year, the brand is facing a reputation crisis. Reach out to dissatisfied Quarter 4 customers with service credits or partial refunds to prevent negative word-of-mouth.
- Stop the "Average" Thinking: Management must stop relying on the "Okay" annual rating of 3.13. The business must shift its focus to the current 2.40 rating to acknowledge that the company is presently in an operational emergency.
- Optimize Regional Inventory: Data shows a "one-size-fits-all" approach is failing. Double down on Chevrolet in Texas and Toyota in Florida, while reserving luxury brands like Audi exclusively for California where the demand actually exists.
- Protect Profit Margins: Net revenue is falling twice as fast as order volume. The company must review its discounting strategy to ensure it isn't over-discounting high-demand models just to maintain sales numbers.
- Incentivize Low-Fee Payments: Since all credit cards currently receive the same ~0.62% discount, the company is losing money on high-fee cards like American Express. Offer slightly better incentives for lower-fee "business-friendly" payment methods.
- Implement a Real-Time Dashboard: The steady four-quarter slide proves that waiting for quarterly reports to make decisions is too slow. The CEO needs weekly visibility on shipping delays and feedback to catch problems before they compound.
- Rebuild the Sales Pipeline: A 35.8% drop in orders shows that new customers are being scared off by online ratings. The company needs a targeted marketing campaign that highlights "Verified Faster Shipping" once logistics are improved.
- Prioritize Customer Retention: With 994 customers and only 1000 orders, there is almost zero repeat business. Introduce a loyalty or referral program to turn one-time buyers into long-term brand advocates.
- Address the Quality Gap: The shift from "Very Good" to "Very Bad" suggests that after-sales care is not meeting expectations. Conduct an audit of the vehicle inspection process to ensure that the "pre-owned" vehicles meet the quality standards promised by the app.

In summary, the data presents a clear mandate for **New-Wheels**: the business is currently in an operational "red zone," where extreme shipping delays of **174 days** have directly triggered a collapse in customer satisfaction and a **52% drop in revenue**. To move from a "survival phase" back to growth, the leadership must pivot from high-volume expansion to **operational stability**, prioritizing logistics speed and customer recovery above all else. Addressing these core service failures is the only path to rebuilding brand trust and securing the company's long-term financial health.

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